TECHNICAL REPORT · PMDC MOTOR CONTROL & ANALYSIS

DriveBench

Speed regulation of a permanent-magnet DC drive with PID, optimised PID, LQR and model predictive control

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Summary

Four controllers regulate the shaft speed of one simulated motor through one shared voltage drive. They see identical references, load torques, bus levels and encoder noise in eleven runs. MPC has the lowest speed-error integral (IAE) in 6 of the 11 runs, LQR in 4, and 1 is a tie (within 0.5 %); LQR is the fastest to settle at startup (0.71 s) and the best at load disturbances (IAE 1.2 against 3.2 for MPC); MPC keeps the drive out of its current limit for less than half the time of the others under overload (4.34 s against 10.05 s for LQR) but costs about 0.6 ms of computation per 10 ms sample instead of 0.002 ms. The LQR and MPC settings were chosen by hand while looking at these runs, so this is a demonstration of mechanisms, not a neutral tuning contest.

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1 The drive and what limits it

A speed drive of this kind turns conveyors, pumps and machine axes. The controller does not touch the motor directly: it requests an armature voltage, and a drive layer decides what the motor actually receives. Figure 1 shows the signal path used for every controller in this report.

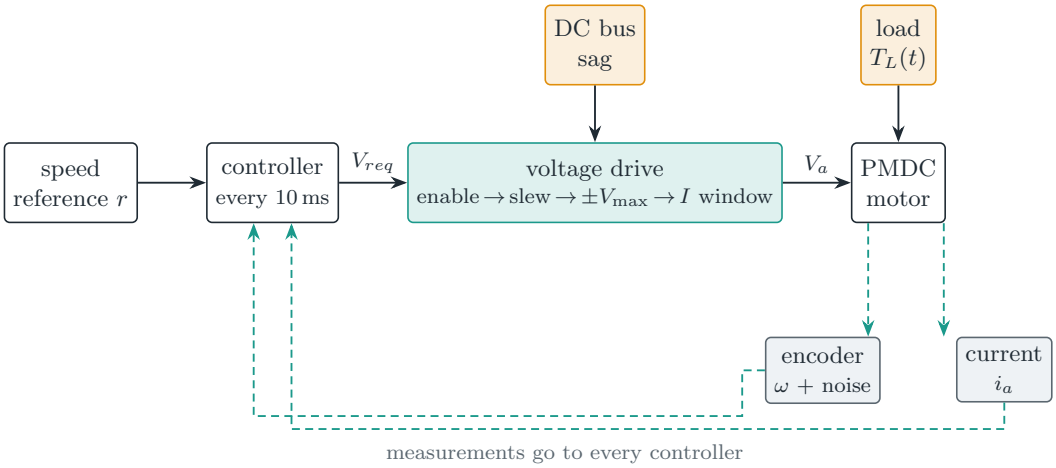


Figure 1. Signal path shared by all four controllers. Only the *controller* block changes between runs; the drive, plant, sensors and disturbances are identical.

The drive applies, in this order: an enable/e-stop gate, a slew limit of $1200\,\mathrm{V\,s^{-1}}$, saturation at $\pm V_{\max}$ times the DC-bus level (24 V nominal), and a current window. The window is not a clip of the simulated current. It restricts the voltage so that the exact one-sample electrical response stays inside $\pm I_{\max}$ (8 A), using the drive's nominal R_a and K_e and the measured current and speed. Under ±20 % winding-resistance error it can be exceeded briefly (up to 9.10 A in the low-resistance corner) before the feedback term pulls the voltage down.

2 The motor model

The plant is the coupled electrical and mechanical system

$$L_a\frac{di_a}{dt} = V_a - R_a i_a - K_e\omega, \qquad J\frac{d\omega}{dt} = K_t i_a - B\omega - T_L, \qquad \frac{d\theta}{dt} = \omega. \tag{1}$$

With the baseline parameters in Table 1 the state matrices for $x = [i_a, \omega, \theta]^\top$ are

$$A = \begin{bmatrix} -100 & -5 & 0 \\ 25 & -0.5 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}, \quad E = \begin{bmatrix} 0 \\ -500 \\ 0 \end{bmatrix}, \tag{2}$$

and the transfer function $\Omega/V_a = K_t/(L_aJs^2 + (L_aB + R_aJ)s + R_aB + K_eK_t)$ has poles at -98.7 and $-1.75\,\mathrm{s^{-1}}$: an electrical time constant of 10 ms and a mechanical one of about 0.57 s. The plant is integrated with RK4 at 1 ms while the controllers update every 10 ms and hold their output between updates.

Table 1. Baseline motor and drive parameters (illustrative, not a commercial motor).

R_a	1.0 Ω	J	$0.002\,\mathrm{kg\,m^2}$
L_a	10 mH	B	$0.001\,\mathrm{N\,m\,s/rad}$
$K_t = K_e$	0.05	nominal T_L	0.05 N m
$V_{\max}$	24 V	$I_{\max}$	8 A
top speed	3000 RPM	slew limit	1200 V/s

Four models, kept apart. The *actual plant* may use mismatched R_a, L_a, J, B. The controllers' *prediction models* are always the nominal motor. The *drive* is a separate layer with its own limits. The *measurement model* adds seeded Gaussian noise to the encoder and measures current ideally.

3 Four control laws

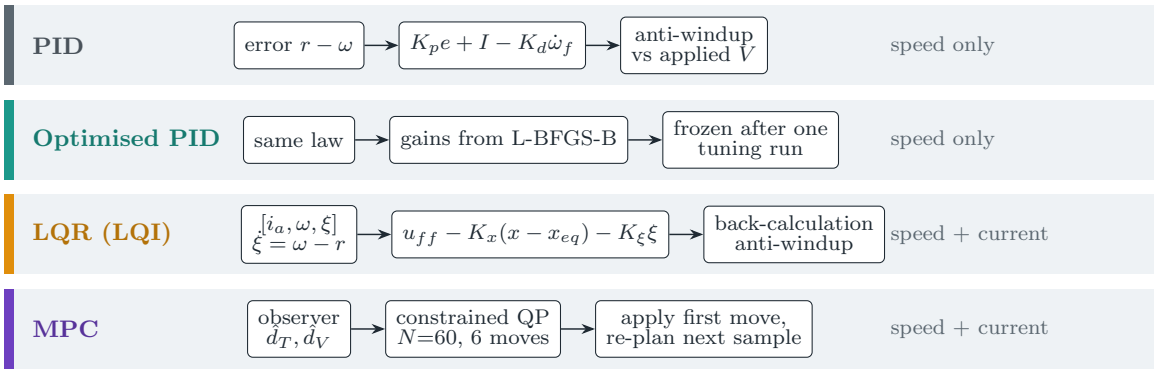


Figure 2. What each controller computes every 10 ms. All four output a voltage request that enters the same drive.

3.1 PID and optimised PID

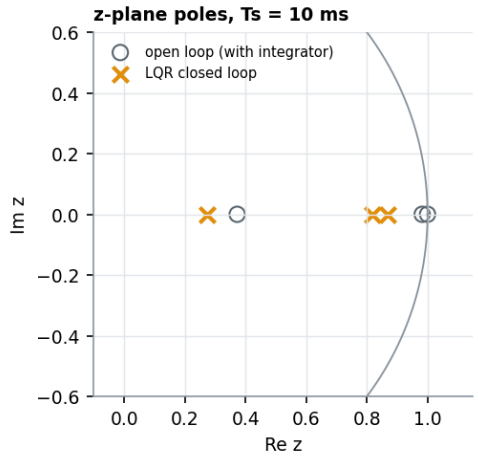
The positional PID takes RPM and returns volts. Its derivative acts on the (low-pass filtered) measurement so reference steps cause no kick [4], and the integrator is unwound by back-calculation against the voltage the drive *applied*, so slew, saturation and current limiting all stop windup. The hand-tuned gains are K_p=0.01 V/RPM, K_i=0.03, K_d=0. The optimised PID uses the same law with gains minimising a weighted sum of IAE, overshoot, settling time, voltage effort and time in voltage saturation and current limit (L-BFGS-B). They were found once on the speed-tracking scenario (K_p=0.03259, K_i=0.1313, K_d=0) and are frozen for every comparison.

3.2 LQR with integral action

Augmenting the state with $\xi = \int(\omega - r)\,dt$ gives a three-state design model. The plant part is discretised exactly (zero-order hold) at $T_s = 10$ ms, and the integrator uses the same forward-Euler update the controller executes, so the Riccati design certifies the loop that actually runs. The cost is $\sum x_{aug}^\top Q x_{aug} + u^\top R u$ on states scaled by $I_{\max}$, $\omega_{\max}$, $0.5\,\mathrm{s}\cdot\omega_{\max}$ and voltage scaled by $V_{\max}$, with dimensionless weights $q_i = 0.1$, $q_\omega = 300$, $q_\xi = 10000$, $r_V = 1$. Solving the discrete algebraic Riccati equation [3] gives the gain

$$K = \left[K_i, K_\omega, K_\xi\right] = \left[0.416,\ 1.461,\ 11.109\right]\ (\mathrm{V/A},\ \mathrm{V/(rad/s)},\ \mathrm{V/rad}), \tag{3}$$

and the law is $u = u_{ff}(r) - K_x(x - x_{eq}(r)) - K_\xi \xi$ with the zero-load equilibrium as feedforward. The design is rejected, and no gain used, when the augmented pair is uncontrollable, the Riccati equation fails or any closed-loop pole leaves the unit circle. Figure 3 shows the result: the electrical pole moves in to about 0.27 and all closed-loop poles lie within $|z| \le 0.868$.



Reading the plot. Circles are the poles of the discretised augmented plant before feedback: the fast electrical pole near $z = 0.37$ and the two slow modes (mechanical and integrator) close to $z = 1$. Crosses are the closed-loop poles. Feedback of current moves the electrical pole inward, and the integral gain pulls the slow pair away from the unit circle, which is what removes steady-state speed error.

Figure 3. LQR pole map for the recommended weights.

3.3 Offset-free MPC

MPC predicts with the two-state model $x_{k+1} = A_d x_k + B_d(V_k + \hat{d}_V) + E_d \hat{d}_T$, $x = [i_a, \omega]^\top$. Each sample the innovations of the measured current and speed are mapped to a load-torque estimate $\hat{d}_T$ and an armature-voltage estimate $\hat{d}_V$ (observer gain 0.1). This follows the offset-free disturbance-model idea of [1] within the framework of [2]. Estimating both matters: with a torque-only estimate the speed kept a 44 RPM offset under +20 % resistance in development. Over a horizon of $N = 60$ samples (0.6 s) it chooses six blocked voltage moves minimising weighted speed error, a terminal speed error and voltage change, subject to the voltage box, the voltage change between moves, and the *predicted* armature current $|i_{a,k}| \le I_{\max}$ over the whole horizon (Figure 4). SciPy SLSQP solves it with analytic gradients because all constraints are linear in the moves.

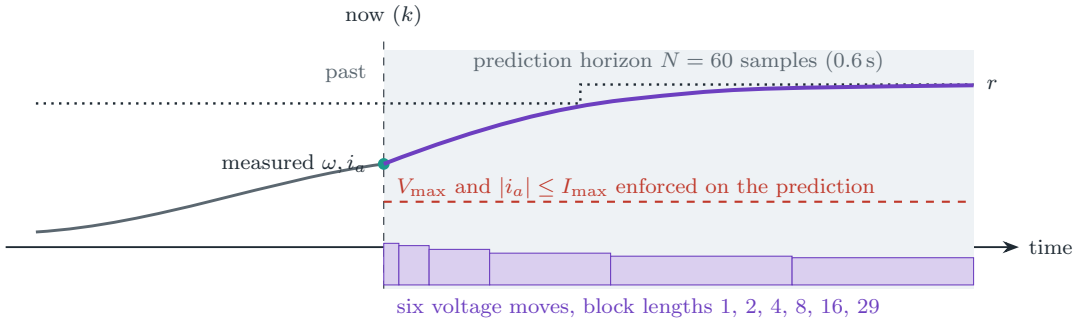


Figure 4. The receding horizon: only the first planned move is applied; the plan is recomputed from the new measurement at the next sample.

4 How the comparison is kept fair

Table 2. What is shared, and what differs.

	Identical for all four	Differs
Plant and drive	motor parameters, drive limits, slew, current window	—
Inputs	reference, load torque, bus level, seeded noise realisation, initial state	—
Timing	10 ms controller, 1 ms plant, zero-order hold	—
Measurements	encoder speed with the same noise	LQR and MPC also use measured current; PID loops do not
Tuning	optimised PID tuned once, frozen	LQR weights and MPC settings hand-set
Cost	same metric code	MPC solves an optimisation each sample

A test asserts that the reference, load and noise arrays are identical across controllers, and that all four stay inside the drive's voltage limit in every scenario.

5 Results

Eleven runs cover startup, four-step tracking, sudden load, load removal, voltage saturation (2900 RPM with a bus sag to 85 %), current protection (a load that needs more than 8 A), model mismatch at both corners (R_a+20%, J+30%, B+25% and the opposite), encoder noise (15 RPM), variable load (0.5 Hz sinusoid plus bounded random torque) and a reversal. Table 3 lists the headline metrics and Figure 5 ranks the error integral. All numbers are produced by the repository's scripts.

Table 3. Error integral, settling time and time in current limit. Shaded cells are the best of the four (no shading when all tie; "n/a" means the speed never stayed inside the 2 % band).

Scenario	IAE [RPM s]				Settling [s]				Current-limited [s]			
	PID	Opt.	LQR	MPC	PID	Opt.	LQR	MPC	PID	Opt.	LQR	MPC
Motor startup	414	364	**346**	351	1.84	0.92	**0.71**	0.76	**0.20**	0.45	0.67	0.60
Speed-reference tracking	1854	1660	1608	**1071**	1.58	1.49	1.46	**0.87**	**1.81**	2.65	3.42	3.05
Sudden mechanical load	35	8	**1**	3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Load removal	35	8	**1**	3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Voltage saturation	4122	4092	4072	**3307**	n/a	n/a	n/a	n/a	**3.00**	3.53	3.98	3.84
Current protection	3678	3635	**3616**	3626	3.92	**3.89**	**3.89**	**3.89**	9.18	9.71	10.05	**4.34**
Model mismatch	1794	1700	1676	**1290**	2.69	2.67	2.67	**2.08**	**4.10**	4.25	4.36	4.47
Sensor noise	181	160	173	**103**	0.68	0.63	0.46	**0.25**	**0.08**	0.22	0.54	0.16
Variable load	275	96	**18**	45	11.74	**0.00**	**0.00**	**0.00**	0.00	0.00	0.00	0.00
Reverse operation	2199	2079	2037	**1132**	2.37	1.44	1.24	**0.69**	**1.10**	1.65	2.10	1.62
Model mismatch (low R, J, B)	756	615	518	**468**	1.17	0.87	0.57	**0.33**	**0.25**	0.50	0.95	0.39

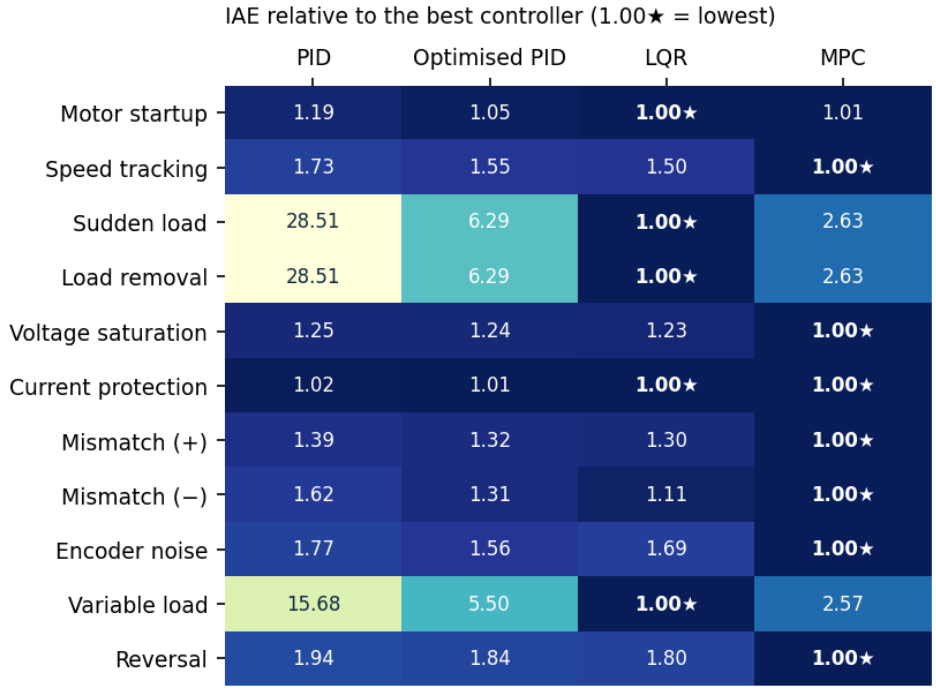


Figure 5. IAE relative to the best controller in each run. MPC and LQR split the wins; the PID variants are far behind on load disturbances.

5.1 Transient shapes

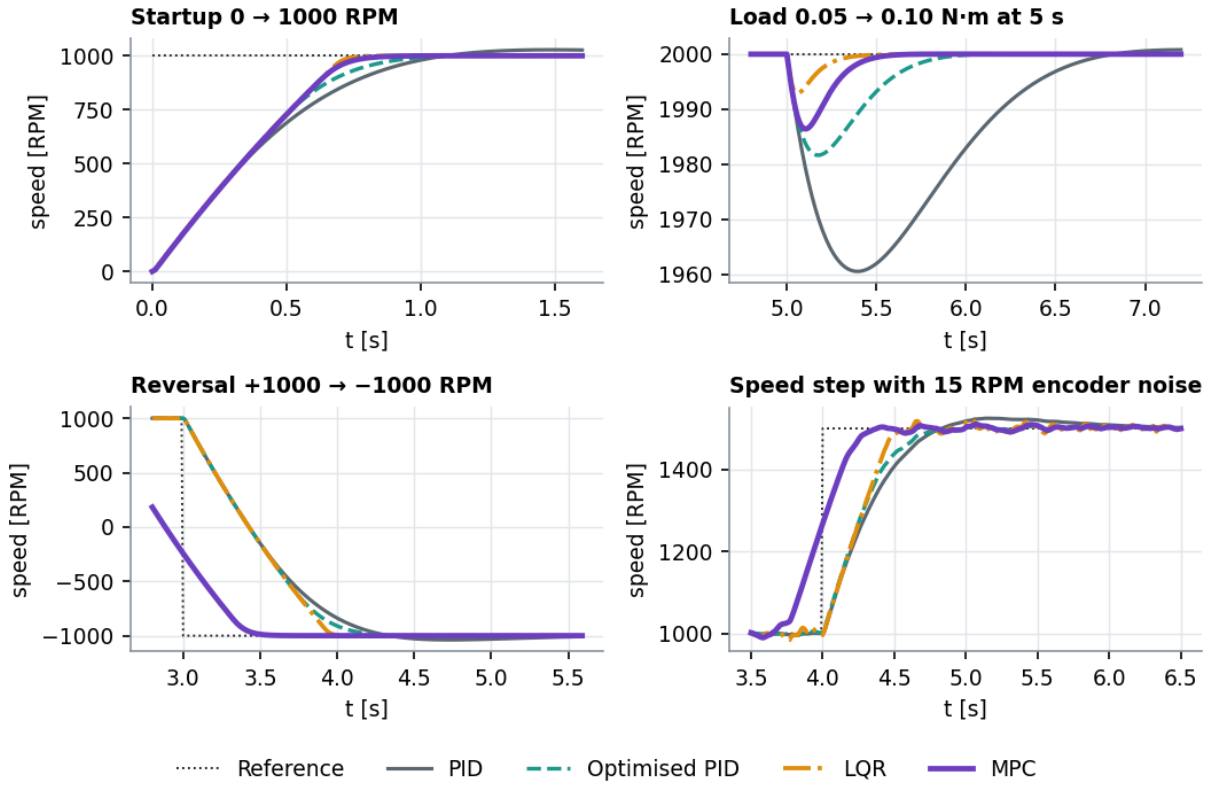


Figure 6. Zoomed speed responses. In the reversal panel the MPC trace leaves 1000 RPM before the reference changes: it previews the upcoming step.

Startup is limited by the 8 A current ceiling for all controllers, so differences come from how late each one leaves the ceiling (Figure 7): LQR holds the current limit longest and settles first (0.71 s against 0.76 s for MPC and 1.84 s for the hand-tuned PID). On the load step the integral-plus-current-feedback structure of the LQR reacts fastest (peak dip well under the others), MPC next through its disturbance estimate, and the PIDs only after the speed error has grown.

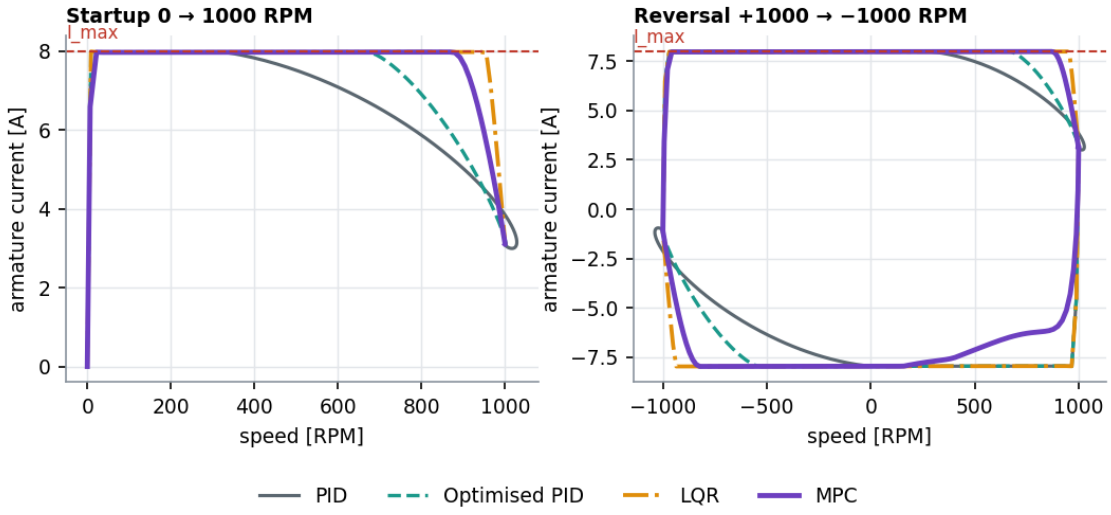


Figure 7. Armature current against speed. The red dashed line is $I_{\max}$.

5.2 Current protection

When the load needs more than 8 A, nothing can hold 2500 RPM, and the question becomes how the drive spends its time at the limit. Figure 8 shows the intervals during which the drive's current window was active. MPC leaves the limit for most of the second half (4.34 s in total), while the other three stay in it (9.18, 9.71 and 10.05 s), because only MPC sees the limit in its prediction. Their error integrals are nonetheless within 2 % of each other (3678, 3635, 3616, 3626), since the actuator, not the algorithm, sets the best achievable speed.

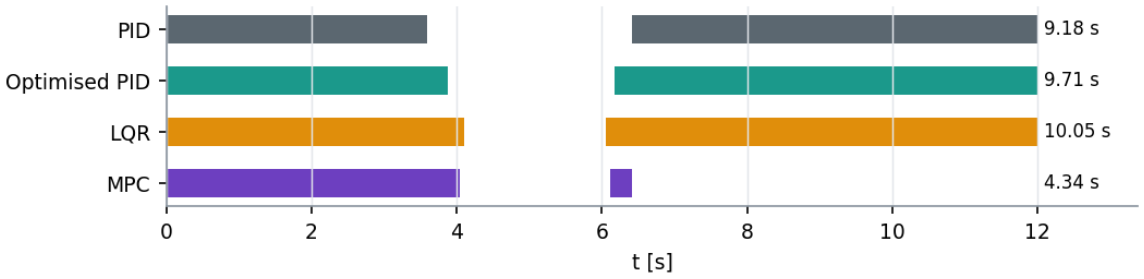


Figure 8. Intervals with the current window active in the current-protection run; the figure on the right is the total time.

5.3 Robustness and noise

Figure 9 compares the nominal, mismatched and noisy runs. Under +20% resistance MPC stays offset-free with its two disturbance estimates (IAE 1290, against 1700 for the frozen optimised PID and 1706 for a PID retuned on the mismatched plant, and 1676 for LQR). With 15 RPM encoder noise the MPC keeps the lowest IAE (103) after the observer gain was lowered to 0.1; at gain 1 it amplified the noise into the voltage and lost to the PID loops.

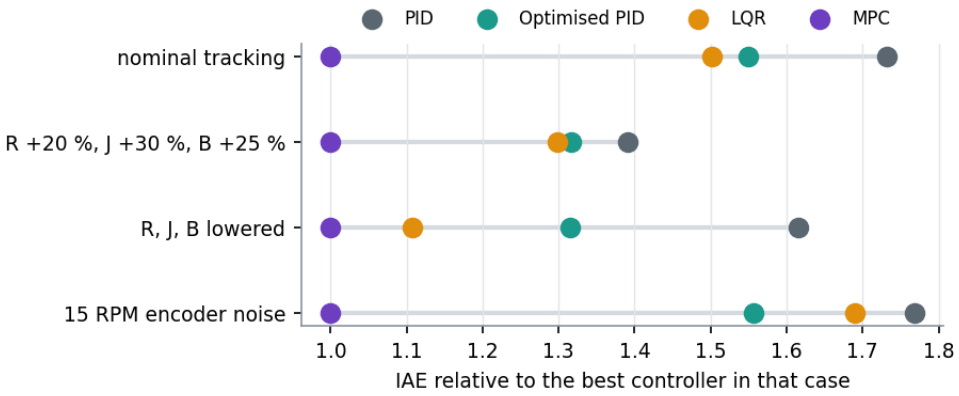
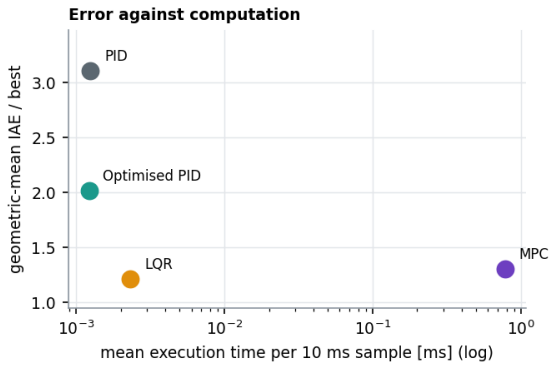


Figure 9. IAE relative to the best controller in four cases. A dot at 1.0 is the best.

5.4 Error against computation



Averaged geometrically over the eleven runs, the LQR's IAE is 1.21 times the best in each run, MPC's is 1.30, the optimised PID's 2.01 and the hand-tuned PID's 3.10. The LQR gets there with about 0.002 ms of computation per sample; MPC needs roughly 0.6 ms, a few hundred times more, and wins where previewing the reference or the current limit matters. The solve is well inside the 10 ms period on this machine, but it is not a hard real-time guarantee.

Figure 10. Geometric-mean IAE ratio against mean execution time (log axis).

6 Limitations

- The LQR weights (chosen from a short sweep on startup) and the MPC horizon, weights and observer gain were set by hand while looking at these runs.
- LQR and MPC use measured current and a nominal model that matches the plant when no mismatch is applied; the PID loops are single speed loops.
- The motor is linear with constant parameters: no brush drop, saturation, temperature drift, Coulomb friction or cogging. The current sensor is ideal and the load keeps its sign at standstill.
- Efficiency figures use an illustrative 0.95 power-stage efficiency and are not data.
- No stability proof is claimed for any controller under saturation, and the MPC solve time is not a real-time guarantee.
- The parameters are illustrative; none of the numbers describe a commercial motor.

7 Reproducing this report

```
python scripts/rebuild_report_metrics.py # metrics, figures, results/
python scripts/rebuild_report_charts.py   # tables and number macros
cd report && latexmk -pdf drivebench_report.tex
```

Numbers in the text are macros generated into `report/numbers.tex`; figures are generated into `report/figures/`. Nothing is typed by hand.

References

[1] G. Pannocchia, J. B. Rawlings, "Disturbance models for offset-free model predictive control," *AIChE J.* 49(2):426–437, 2003.

[2] J. B. Rawlings, D. Q. Mayne, M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., Nob Hill, 2017.

[3] B. D. O. Anderson, J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Prentice Hall, 1990.

[4] K. J. Åström, T. Hägglund, *Advanced PID Control*, ISA, 2006.